

4.5 Curve Sketching (Page 315 of the text book)

We have been gathering many tools that can assist us with the sketching of the graphs of functions - Let's list these:

For the function:

- D: We can find the Domain of the function (For which x can $f(x)$ be calculated?)
- I: We can find the intercepts with the axes (Put $x = 0$ to find the y-intercept; $y = 0$ for the x-intercept)
- S: Determine Symmetry (even functions are symmetric about the y-axis; odd functions are symmetric about the origin)
- A: Asymptotes: Find horizontal and vertical asymptotes (use limits for this!)

The acronym DISA can help you to remember the points to consider for the function.

For the derivative f' of f :

- D: Decreasing/increasing (Use the first derivative to find the critical points and the intervals!)
- E: Extremes (Check if the sign (+ or -) of the value of the first derivative changes at a critical point)

For the second derivative f'' of f :

- C: Concavity (Use the second derivative to find the intervals)
- I: Inflection points (find the points where the sign of the concavity changes)

The acronym DECI can help you to remember the points to consider for the first and second derivative.

Example 1 Sketch the graph of the function f with $f(x) = \frac{x^2}{x^2-4}$

Solution:

D: Domain: Clearly we must have $x^2 - 4 = (x - 2)(x + 2) \neq 0$. Thus, $x \neq 2$ and $x \neq -2$.

Thus the domain is $\mathbb{R}/\{-2, 2\}$

I: Intercepts: If $x = 0$, $f(x) = 0$, so the y-intercept is at $y = 0$.

If $y = 0$, $\frac{x^2}{x^2-4} = 0 \Rightarrow x = 0$, so the x-intercept is at $x = 0$.

S: Symmetry: Since $f(-x) = \frac{(-x)^2}{(-x)^2-4} = \frac{x^2}{x^2-4} = f(x)$, the function is even and so we know the graph will be symmetric about the y-axis.

A: Asymptotes: Since the numbers -2 and 2 are not in the domain of f we check these points for vertical asymptotes:

$\lim_{x \rightarrow 2^-} \frac{x^2}{x^2-4} = -\infty$ and $\lim_{x \rightarrow 2^+} \frac{x^2}{x^2-4} = \infty$. Thus, $x = 2$ is a vertical asymptote.

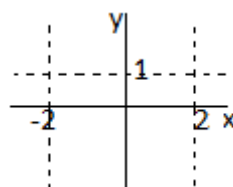
$\lim_{x \rightarrow -2^-} \frac{x^2}{x^2-4} = \infty$ and $\lim_{x \rightarrow -2^+} \frac{x^2}{x^2-4} = -\infty$. Thus, $x = -2$ is a vertical asymptote.

Horizontal asymptotes:

$\lim_{x \rightarrow \infty} \frac{x^2}{x^2-4} = \lim_{x \rightarrow \infty} \frac{1}{1-1/x^2} = 1$ and $\lim_{x \rightarrow -\infty} \frac{x^2}{x^2-4} = \lim_{x \rightarrow -\infty} \frac{1}{1-1/x^2} = 1$

Thus, $y = 1$ is a horizontal asymptote.

At this stage you can already draw a rough sketch, like this:



D: Decreasing/Increasing: $f'(x) = \frac{(x^2-4)(2x)-x^2(2x)}{(x^2-4)^2} = -\frac{8x}{(x^2-4)^2}$

It follows that $f'(x) = 0 \Leftrightarrow x = 0$. It is also true that $f'(x)$ does not exist when $x = -2$ or $x = 2$, but since these numbers are not in the domain of f the only critical number is $x = 0$.

It also follows that $f'(x) > 0$ if $x < 0, x \neq -2$ and $f'(x) < 0$ if $x > 0, x \neq 2$. Thus, f is increasing on $(-\infty, -2)$ and on $(-2, 0)$ and decreasing on $(0, 2)$ and $(2, \infty)$.

E: Extremes: Since the sign of $f'(x)$ changes from positive to negative at $x = 0$, f has a local maximum of $f(0) = 0$ at $x = 0$.

C: Concavity: $f''(x) = \frac{(x^2-4)^2(-8) - (-8x)(2)(x^2-4)2x}{(x^2-4)^4} = \frac{24x^2+32}{(x^2-4)^3}$

It is clear that $24x^2 + 32 > 0$ for all $x \in D_f$ and so the sign of $f''(x)$ depends on the sign of the denominator

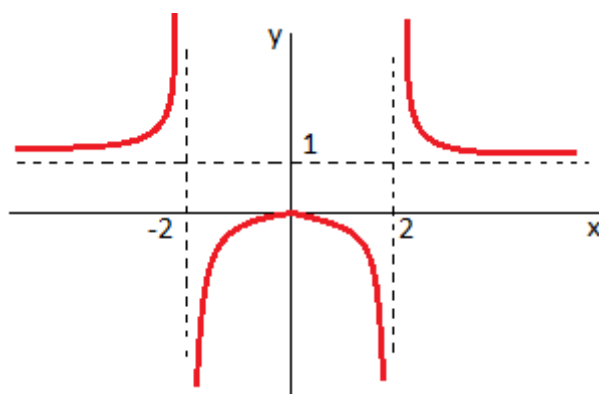
$(x^2 - 4)^3 = (x^2 - 4)^2(x^2 - 4)$. Since $(x^2 - 4)^2 > 0$ for all $x \in D_f$, the sign of $f''(x)$ depends only on the sign of $x^2 - 4$. You can easily sketch the graph of the parabola $y = x^2 - 4$ to find that

$f''(x) < 0$ if $-2 < x < 2$ and $f''(x) > 0$ if $x < -2$ or $x > 2$.

Thus, f is concave upward on $(-\infty, -2)$ and on $(2, \infty)$; f is concave downward on $(-2, 2)$.

I: Inflection points: The value of $f''(x)$ changes from positive to negative at $x = -2$ and changes from negative to positive at $x = 2$. However, since the numbers -2 and 2 are not in the domain of f , there are no inflection points.

Using all of the information we have gathered, we can sketch the graph:



Example 2 Sketch the graph of the function f with $f(x) = xe^{-3x}$

D: Domain: $D_f = \mathbb{R}$

I: Intercepts: If $x = 0$, $y = f(x) = 0$, so the y -intercept is $y = 1$. Also $f(x) = 0 \Rightarrow x = 0$, and so the x -intercept is $x = 0$.

S: Symmetry: The function is neither even nor odd.

A: Asymptotes: Since the domain of f is $\mathbb{R}$ we need only check for horizontal asymptotes:

$$\lim_{x \rightarrow \infty} xe^{-3x} (0 \cdot \infty) = \lim_{x \rightarrow \infty} \frac{x}{e^{3x}} \left(\frac{\infty}{\infty} \right) = \lim_{x \rightarrow \infty} \frac{1}{3e^{3x}} \text{ (L'Hospital's rule) } = 0$$

$$\lim_{x \rightarrow -\infty} xe^{-3x} (0 \cdot \infty) = \lim_{x \rightarrow -\infty} \frac{x}{e^{3x}} \left(\frac{\infty}{\infty} \right) = \lim_{x \rightarrow -\infty} \frac{1}{3e^{3x}} \text{ (L'Hospital's rule) } = -\infty$$

Thus, $y = 0$ is a horizontal asymptote.

D: Decreasing/Increasing: $f'(x) = e^{-3x} + x(-3)e^{-3x} = e^{-3x}(1 - 3x)$

It follows that $f'(x) = 0 \Leftrightarrow x = 1/3$. It is also true that $f'(x)$ exist for all $x \in \mathbb{R}$ and so the only critical number is $x = 1/3$.

Since $e^{-3x} > 0$ for all $x \in D_f$, it follows that $f'(x) > 0$ if $1 - 3x > 0$, that is $f'(x) > 0$ if $x < 1/3$

AND $f'(x) < 0$ if $1 - 3x < 0$, that is $f'(x) < 0$ if $x > 1/3$.

Thus, f is increasing on $(-\infty, 1/3)$ and decreasing on $(1/3, \infty)$.

E: Extremes: Since the sign of $f'(x)$ changes from positive to negative at $x = 1/3$, f has a local maximum of

$$f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)e^{-3(1/3)} = \frac{1}{3e} \text{ at } x = \frac{1}{3}.$$

C: Concavity: $f''(x) = (-3e^{-3x})(1 - 3x) + e^{-3x}(-3) = 3e^{-3x}(3x - 2)$

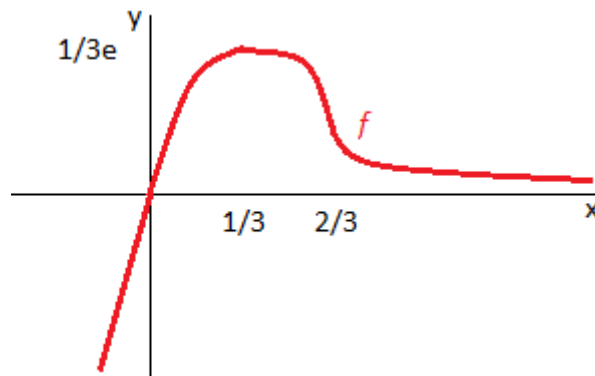
It is clear that $e^{-3x} > 0$ for all $x \in D_f$ and so the sign of $f''(x)$ depends on the sign of $(3x - 2)$.

Thus, $f''(x) > 0$ if $x > 2/3$ and $f''(x) < 0$ if $x < 2/3$.

Thus, f is concave downward on $(-\infty, 2/3)$ and f is concave upward on $(2/3, \infty)$.

I: Inflection points: The value of $f''(x)$ changes from negative to positive at $x = 2/3$ and so f has an inflection point at $x = 2/3$.

Using all of the information we have gathered, we can sketch the graph:



Example 3: Sketch the graph of the function f with $f(x) = 12x^{1/2} - 2x^{3/2}$

HOMEWORK!